Ayan Ahmed

Software Engineer, Google

H +91 9035212894
B, monishaj.65@gmail.com
Í monisha-jega.githu
monisha-jega
monisha-jegadeesan

Education

2015-2020 **Dual Degree (B.Tech + M.Tech) in Computer Science and Engineering**
Indian Institute of Technology Madras, Chennai, India CGPA: 8.78

2015 **XII - Karnataka Board,** *KLE Society's Independent PU College, Bangalore* 97.30 %

2013 **X - ICSE,** *B P Indian Public School, Bangalore* 96.33%

Professional Experience

Dec 2022 - Present **Software Engineer, Level IV, Google LLC, New York**
- Working on Keep, a notetaking editor in Google Workspace.

Aug 2020 - Nov 2022 **Software Engineer, Level IV, Google India Pvt Ltd, Bangalore**
- Developing intelligent features for the Google Workspace Editors (Docs, Slides, Keep, etc) using my expertise on the products' client-side software, supporting tools and libraries, and natural language processing infrastructure.
- Using cutting-edge frontend tools like Web Assembly and Emscripten, and Google-internal technologies like j2Cl, client-side cross-platform frameworks and build systems, to develop user-facing features such as spellcheck in encrypted documents for five languages and writing style suggestions for English text.
- Formulating technical designs for independent end-to-end problems, driving cross-team collaboration, upholding software reliability practices, technical-debt resolution and documentation, and proactively identifying areas of future work.
- Guiding junior engineers on programming and software design tasks to enable timely delivery of products to customers.

May 2019 - July 2019 **Software Engineering Intern, Google India Pvt Ltd, Bangalore**
Worked on the Editors client-side software infrastructure to develop a user interface with control options to undo or provide feedback on the correction and a logging framework, for the Google Docs text auto-correction feature.

May 2018 - July 2018 **Research Intern, Big Data Experience Labs, Adobe Research, Bangalore**
Developed a mobile application for Text to Scene Conversion in Augmented Reality, based on novel research techniques for prediction of three-dimensional object sizes and positions from textual features.

Research Experience

Sep 2019 - May 2020 **Paraphrase Generation with a Bilingual Model and Continuous Embeddings** *Master's Thesis,* ***Language Technologies Institute, Carnegie Mellon University***
Machinated a novel technique for paraphrase generation using the von Mises-Fisher (vMF) Loss on a transformer network, and showed that it produces superior paraphrases as compared to the log-likelihood model by employing bilingual data to induce zero-shot paraphrasing, guided by *Prof. Yulia Tsvetkov*.

May 2017 - July 2017 **Cognitive Approach to Natural Language Processing**
Research Intern, ***Department of Computer Science and Automation, Indian Institute of Science (IISc), Bangalore***
Developed a cognitive text parser that combines syntactic and semantic approaches, to process textual data into cognitive structural representations, to be used as a feature extractor for downstream NLP tasks, and demonstrated the correlation of the extracted cognitive features with semantic and syntactic text features, guided by Prof. Veni Madhavan.

Publications and Patents

[Publication and Poster] **Improving the Diversity of Unsupervised Paraphrasing with Embedding Outputs (Paper, Poster)**
Monisha Jegadeesan*, Sachin Kumar, John Wieting, Yulia Tsvetkov*
In **Workshop on Multilingual Representation Learning**,
The 2021 Conference on Empirical Methods in Natural Language Processing (**EMNLP 2021**)

[Publication and Poster] **Adversarial Demotion of Gender Bias in Natural Language Generation (Paper, Poster)**
Monisha Jegadeesan
In **ACM CODS-COMAD 2020** - Young Researchers' Symposium

[Poster] **ARComposer: Authoring Augmented Reality Experiences through Text (Poster)**
Sumit Kumar, Paridhi Maheshwari, ***Monisha Jegadeesan****, Amrit Singhal, Kush Kumar Singh, Kundan Krishna*
In ACM User Interface Software and Technology Symposium 2019 (**ACM UIST 2019**)

[Filed Patent] **Visualizing Natural Language through 3D Scenes in Augmented Reality**
Sumit Kumar, Paridhi Maheshwari, ***Monisha Jegadeesan****, Amrit Singhal, Kush Kumar Singh, Kundan Krishna*
Filed at the US PTO (Application Number: 16/247,235)

[Publication and Poster] **Leveraging Ontological Knowledge for Neural Language Models (Paper, Poster)** *Ameet Deshpande, **Monisha Jegadeesan***
In **ACM CODS-COMAD 2019** - Young Researchers' Symposium

Projects

Real-Time AI Chatbot for Enterprise Customer Support (2024)

Developed and deployed a scalable AI-powered chatbot leveraging LangChain and OpenAI API, integrated with a vector database built on ChromaDB to enable efficient retrieval-augmented generation (RAG) for instant, accurate customer query responses. Optimized the inference pipeline for reliability and performance, significantly reducing human agent workload by 35% and enhancing response speed. This project involved cloud-based deployment and API orchestration to support real-time LLM inference services, aligning with enterprise-level AI platform standards.

Tech: LangChain, OpenAI API, ChromaDB, Python, LLM Integration, Kubernetes

Customer Sentiment Analysis Platform for E-commerce (2024)

Engineered a robust NLP pipeline to classify customer sentiment from social media data using Python, NLTK, and TF-IDF feature extraction, followed by a Logistic Regression model. The system improved insights into customer feedback, identifying key themes to enhance product offerings. The solution emphasized scalability and API integration for streamlined inference within a cloud infrastructure, ensuring reliable and high-performance NLP service delivery consistent with modern AI deployment practices.

Tech: Python, NLTK, TF-IDF, Logistic Regression, Cloud API Integration

Predictive Maintenance AI Model with Industrial IoT Integration (2024)

Built and operated a large-scale predictive maintenance model using Random Forest on time-series sensor data from industrial IoT devices, focusing on reliability and real-time inference scalability. Designed feature extraction pipelines optimized for high throughput and deployed the model on a cloud-based Kubernetes platform to ensure resilient and performant AI service delivery. The system reduced unexpected machine downtime by 28%, translating into significant operational cost savings and demonstrating strong orchestration of AI inference pipelines in manufacturing environments.

Tech: Python, Random Forest, Industrial IoT, Time-Series Modeling, Kubernetes, Cloud Infrastructure

Telecom Customer Churn Prediction System with Scalable AI Inference (2024)

Developed an advanced machine learning pipeline to predict telecom customer churn using Python and XGBoost, incorporating extensive feature engineering, data preprocessing, and SMOTE for imbalanced dataset handling. The model was deployed via scalable APIs on cloud infrastructure orchestrated with Kubernetes, enabling real-time inference for proactive retention strategies. Achieved a ROC-AUC score of 0.87 and enabled targeted customer engagement that reduced churn by 18%, illustrating integration and operational excellence in AI inference platforms for enterprise-level applications.

Tech: Python, Pandas, NumPy, Scikit-learn, XGBoost, SMOTE, Kubernetes, Cloud API

About :

I am an AI Engineer specializing in building and optimizing intelligent systems leveraging large language models (LLMs), with a strong focus on Kubernetes-based orchestration and cloud infrastructure. Experienced in developing scalable Retrieval-Augmented Generation (RAG) systems, chatbot architectures, and AI-driven applications using LangChain, OpenAI APIs, and ChromaDB. Skilled at integrating, operating, and optimizing LLM inference APIs to ensure reliability, scalability, and performance in enterprise and government AI solutions. Committed to delivering robust, high-impact AI platforms by collaborating with cross-functional teams and deploying solutions optimized for real-world, large-scale inference environments.